INTERIM FEDERAL SPECIFICATION

BEANS WITH PORK, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the U.S. Army Natick Research and Development Laboratories, Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers beans with pork thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, and other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

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Military Specification:

MIL-L-35078 - Loads, Unit, Preparation of Nonperishable Subsistence Items; General Specification For

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Canned Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services (USDA)

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 100-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Food Quality Assurance Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

United States Standards for Beans

(Application for copies should be addressed to the Management Division, Federal Grain and Inspection Service, U.S. Department of Agriculture, Washington, DC 20250.)

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United States Standards for Grades of Canned Pork and Beans

United States Standards for Grades of Tomato Paste

United States Standards for Grades of Sugarcane Molasses

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1, and 6.4).

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3.2 Material. All material shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-flavors, off-odors, and off-colors.

3.2.1 Beans. The beans shall be dry, mature pea beans or small white beans (excluding soy beans) that conform to U.S. No. 1 grade or higher of the U.S. Standards for Beans.

3.2.2 Pork. Pork shall be of the type customarily used in the preparation of products known as beans in tomato sauce with pork or beans with pork and shall be prepared in an establishment regularly operated under supervision of, and in accordance with, the regulations of the Meat and Poultry Inspection Operations, Food Safety and Inspection Service, U.S. Department of Agriculture.

3.2.3 Tomato paste. Tomato paste shall meet the requirements for U.S. Grade A of the United States Standards for Grades of Tomato Paste. The tomato paste shall have a natural tomato soluble solids content of 24 percent or more but less than 28 percent.

3.2.4 Molasses, sugarcane. The sugarcane molasses shall meet the requirements for U.S. Grade A of the United States Standards for Grades of Sugarcane Molasses.

3.2.5 Sugar, brown. Brown sugar shall have been produced from sugar cane or sugar beets, shall possess an excellent flavor, and shall be dark brown in color. The sugar shall comply with the bacteriological requirements of MIL-STD-900.

3.2.6 Onion powder. Onion powder shall have been prepared from white varieties of onion which have prominent pungent characteristics and do not yield a bitter or mild dehydrated product. The product shall be free flowing and practically free from black or off-colored specks, outer skin, outer root base, seed stems and harmless extraneous vegetable material. One ounce of onion powder when stirred into 2 fluid ounces of cool water, 75 $\pm$ 5°F, shall rehydrate to a product with a uniform white color and a characteristic pungent onion flavor and odor, free from any objectionable flavor or odor.

3.2.7 Salt. Salt shall be clean and free from extraneous material. Commonly used recrystallized or vacuum pan granulated salt, with or without anti-caking agent, will be acceptable for this purpose. Iodized salt shall not be used.

3.2.8 Cornstarch. Cornstarch shall be a white, odorless, finely pulverized, modified waxy maize food starch for use in canned foods. The starch shall be bland with essentially no cereal or starchy taste and shall comply with the bacteriological requirements of MIL-STD-900.

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3.2.9 Mustard flour. Mustard flour shall be a bright yellow powder. The mustard flour shall be prepared from a blend of powder derived from milling the endosperm or interior portion of the seed of Brassica hirta Moench and the whole mustard (after removal of the outer cover hulls or husk and with or without the removal of a portion of the fixed oil) of Brassica juncea. Volatile oil content shall be not less than 0.4 milliliters per 100 grams.

3.2.10 Garlic powder. Garlic shall be dehydrated garlic powder that is free-flowing, uniformly light in color, pungent, and practically free from outer skin and scorched particles. The powder shall be free from husk, root base, stem, soil, and mold.

3.2.11 White pepper. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or outer and inner coverings have been removed. Volatile oil content shall not be less than 1 milliliter per 100 grams of pepper.

3.3 Bean preparation. The beans shall be thoroughly cleaned to remove dirt, stones, loose skins, and any other extraneous material and sorted to remove those which are spotted, discolored or affected by pathological or insect injury. The beans shall be thoroughly washed, soaked to full absorption (50 - 60 percent moisture) and shall be blanched prior to the canning process.

3.4 Sauce preparation. The sauce shall be prepared from the following ingredients:

Ingredient	Percent by weight
Water	61.32
Pork	13.74
Tomato paste (24 percent or more but less than 28 percent natural tomato soluble solids)	10.90
Molasses	5.50
Brown sugar	3.60
Onion powder	3.00
Salt	1.75
Cornstarch	0.10
Mustard flour	0.05
Garlic powder	0.03
White pepper	0.01
	100.00

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The pork shall be sliced and cut transversely into 1-1/2 to 2 inch strips, sauteed, and added with the drippings to the homogenous slurry prepared from the remaining ingredients. The mixture shall be heated to 212°F and held for a period of 5 minutes at that temperature. The sauce shall be used within a 4-hour period after preparation and shall be maintained at 150 ± 10°F during this period. The sauce shall be thoroughly mixed.

3.5 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with approximately 58 ounces of beans and 48 ounces of sauce such as to conform to the following requirements:

a. The temperature of beans and sauce at time of filling shall be not less than 150°F.

b. The average net weight per can shall be 106 ounces minimum.

c. No individual can shall contain less than 104 ounces of product.

d. The product net weight to drained weight ratio shall be 1.59 ± 0.18.

e. The filled can shall be hermetically sealed.

3.6 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermoprocessed by retorting at 240°F until a sterilization value (F_o) of not less than 6.0 has been achieved at the coldest heating point of the retort and the product has been thoroughly cooked without overcooking.

3.7 Finished product requirements. The tray pack beans with pork shall meet the requirements for pea beans or small white beans and pork in tomato sauce, U.S. Grade A of the United States Standards for Grades of Canned Pork and Beans. The salt content of the finished product shall not be less than 1.5 nor more than 2.0 percent.

3.8 Plant qualification. The product shall be prepared, processed and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable Government inspection agency.

3.9 Federal Food, Drug and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation and related services shall be borne by the contractor.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected for conformance to the characteristics indicated in 3.7, 4.7.1 4.7.2 and 4.7.3. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity or condition requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.6 In-process examination. In-process examination shall be performed to determine conformance to the ingredient preparation, product processing, can interior coating, and sealing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation; certificates of conformance; or other valid documents shall be cause for rejection of the involved product.

4.7 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

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4.7.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 0.1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.7.2 Drained weight. Eight tray pack cans shall be selected at random from the lot. Each of the eight tray packs shall be opened and inspected for drained weight in accordance with the following procedure. The contents of a tray pack shall be poured into a 12-inch diameter U.S. Standard No. 8 sieve such as to distribute the product evenly in the sieve. The sieve and contents shall be immersed in 70 $\pm$ 2°F water, fat particles shall be removed and the sieve shall be agitated with a vigorous swirling motion for 1 minute taking care to separate any chunks of product into individual beans. The sieve and contents shall be removed from the water and then reimmersed momentarily twice in succession. The sieve shall be tilted slightly and allowed to drain 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The net weight to drained weight ratio shall be calculated to the nearest 0.01. If the ratio for any can is less than 1.41 or more than 1.77 the lot shall be rejected.

4.7.3 Salt content testing. Nine tray packs shall be selected from the lot and distributed as follows:

-Three for laboratory analysis
-Three for submission to the contractor
-Three for retention by the inspector as a reserve sample

The three tray packs for laboratory analysis shall be tested for salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, except that preparation of the sample shall be as follows. The three unopened tray packs shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The lot shall be rejected if the salt content in any individual can is less than 1.5 or more than 2.0. Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the salt content requirement. When the reserve samples are analyzed, the analyses for salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.8 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers.

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4.9 Shipping container examination. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in construction, closure, and reinforcement shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) When applicable, reinforcement with nonmetallic strapping or tape is not used. Level C shipping containers shall be examined only for the marking defects heretofore specified.

4.10 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of the food product, sealed, and thermoprocessed as specified in 3.5 and 3.6. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient weight and temper to protect the product during shipment and storage. The entire inside area of the can shall be coated. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be full length and width less approximately 1/4 inch in each dimension. The box liner and pads shall be fabricated of grade V3c fiberboard.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

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5.2.2.1 Nonmetallic reinforcement of shipping containers. When specified (see 6.1 and 6.3), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. For distribution to Navy, when unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled to comply with the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder. Paper labels are not permitted. The name of the product or abbreviation thereof and date of pack (day, month, year) shall be embossed or lithographed on the end (lid) of the can. Cans shall have the following heating directions and yield.

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 15 minutes. Remove from water and open lid for serving.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 15 minutes until contents reach an internal temperature of 165°F.

YIELD

Approximately 25 servings of 4 ounces each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.3 and 6.4).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) When nonmetallic reinforcement is required for level B packing (see 5.2.2.1).
(e) Type and class of unit load when unit loading is required (see 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 Selection of nonmetallic reinforcement. The nonmetallic reinforcement specified in 5.2.2.1 should be selected for shipping containers that are destined for transfer at sea or other specific overseas operations.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of 4 tray packs of product. The contracting officer should include specific instructions in all acquistion documents regarding arrangements for inspection and approval of the first article.

Preparing activity:

Army - GL

Project No. 8915-A655

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